

Experiencing EmbedNet: Embedding self-sensing to 3D casting objects

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Abstract

This paper introduces EmbedNet, a method for integrating dense sensor networks into casting objects. With EmbedNet, sensor nodes are seamlessly incorporated into casting objects during fabrication. The process involves extruding base materials like silicone rubber or liquid plastic and a custom-designed sensor strip using a hand-held extruder into a mold tailored to specific applications. The base material mixes with the sensor strip in the mold, and upon curing, the result is an object with a defined shape housing a sensor network. EmbedNet employs a small Host node to access sensor data from all nodes on the strip. Each sensor node is self-contained and provides status indications through an on-board RGB LED. The Host connects with all sensor nodes using just three wires: power, ground, and data. This one-wire communication is facilitated through a custom-designed software serial port for each sensor node. The paper showcases various applications of EmbedNet, including wearables, home sensing, and entertainment devices.

Keywords

Sensor network, DIY, self-sensing, soft electronics. 3D electronics

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1 Introduction

We envision a future where creating any 3D object with densely embedded sensors becomes effortless. These sensors seamlessly integrate with the material, enabling measurement of various parameters such as mechanical strain within different parts and environments of the object, and interactions like touch and squeezing. Presently, the addition of electronics is primarily surface-mounted onto the material [1, 4, 5], typically after the fabrication process. Material-embedded sensors are predominantly limited to 2D materials like stretchable silicone tattoos [8, 9] or hard materials like 3D printed electronics [7].

This paper explores the crafting of a self-sensing material capable of being molded into arbitrary shapes. EmbedNet draws inspiration from pioneering concepts such as Radical Atoms [6], Claytronics [3], and Programmable Matter [2], which envision materials with inherent information-processing capabilities. To realize this vision, we introduce *EmbedNet*: a system comprising miniature sensor nodes and an extruder designed to facilitate the casting of sensor networks within objects.

2 Design

The EmbedNet system consists of three main components: a sensor strip, a hand-held extruder, and a base material extruder.

2.1 Sensor strip hardware

The EmbedNet sensor strip comprises multiple wired sensor nodes, accompanied by a Host unit responsible for supplying power and controlling all nodes along the strip, as depicted in Figure 1.

The structural diagram of the entire sensor strip is illustrated in Figure 2.

Each sensor node is equipped with an MCU (STM32C011F6U6) to manage a diverse array of sensors and communications. These sensors include an accelerometer (BMA400), an ambient light sensor (VEML6030) - also called Lux sensor, a barometer with an integrated temperature sensor (DPS310), and an RGB LED. The dimensions of the node are 10 mm x 9.5 mm x 2 mm, as depicted in Figure. 3.

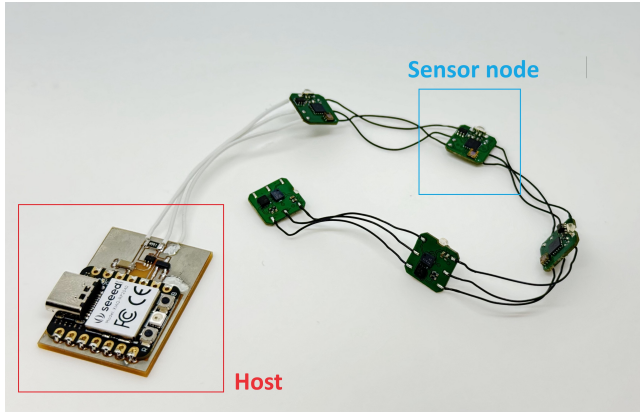


Figure 1: Hardware system of the EmbedNet.

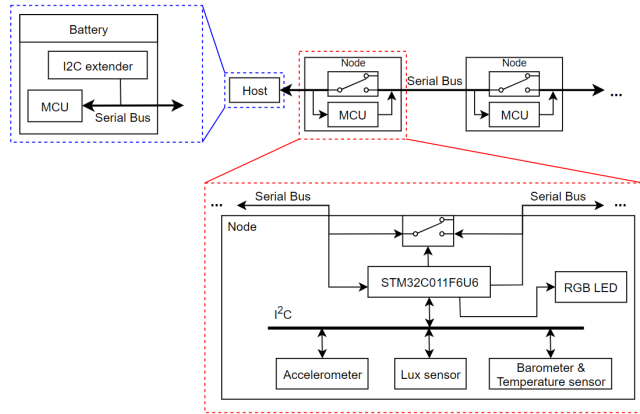


Figure 2: System diagram of the EmbedNet hardware.

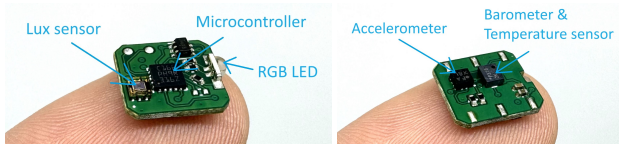


Figure 3: EmbedNet sensor node on a fingertip: (Left) front side, and (right) backside.

The Host unit is outfitted with a battery for power supply, an MCU module responsible for controlling all sensor nodes, facilitating communication between the sensor strip and a computer. All sensor nodes are linked to the Host via a wired connection consisting of three wires: power, ground, and a communication line. The Host utilizes the UART (Universal Asynchronous Receiver / Transmitter) peripheral for communication, while each node is equipped with a custom-designed software UART for data transmission. In the communication paradigm, the Host functions as a client, while each node operates as a server. The Host initiates every data exchange, and the nodes respond to the Host's commands.

2.2 Extruder system design

The EmbedNet extruder system is composed of two parts: a base material extruder to extrude base material like silicone rubber (e.g., Eco-Flex, OOMOO) and liquid plastic (e.g., Smooth-Cast), and a hand-held extruder. These two extruders are connected through a plastic tube. The base material extruder comprises two stepper motors and a 100ml syringe. These stepper motors exert the necessary pushing force to extrude the base material into the mold effectively through the plastic tube. The hand-held extruder serves as the core component of the system and is tasked with extruding the sensor strip and guiding the base material from the plastic tube alongside the sensor strip into a mold. Users can grip and manipulate the extrusion of the base material and sensor strip independently to accommodate various terrain features of different molds. The control of both the sensor strip and the base material extrusion is managed individually through two separate buttons on the hand-held extruder. Figure 4 illustrates the configurations of the extruder.

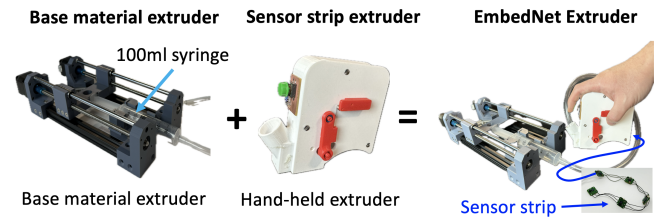


Figure 4: The EmbedNet extruder is composed of a base material extruder and a hand-held extruder. They are connected through a plastic tube.

2.3 Manufacturing process

The manufacturing process utilizing EmbedNet initiates with the design of molds tailored to specific requirements. For materials characterized as soft or deformable, a 3D-printed mold is employed, and subsequently removed post-curing. Conversely, for rigid materials such as liquid plastic, the mold necessitates soft materials like silicone rubber to facilitate easy removal of the rigid object post-curing. The extruder is utilized to fill the mold with the chosen base material and sensor strip. Upon completion of the curing process, the outcome is an object featuring a predetermined shape, material composition, and integrated sensing capabilities.

In the rigid casting process, EmbedNet embeds sensing capabilities into objects cast with rigid materials such as liquid plastic. This process ensures that the sensor nodes are integrated seamlessly into the final object during the casting process, enabling the object to possess both structural integrity and sensing capabilities.

Conversely, in the soft/deformable casting process, EmbedNet adds sensing into flexible materials during the casting process. This approach allows for the creation of objects with deformable properties while retaining the ability to sense various environmental parameters.

3 Applications

The EmbedNet system is versatile, accommodating various base materials, including both rigid and flexible options. Here, we present

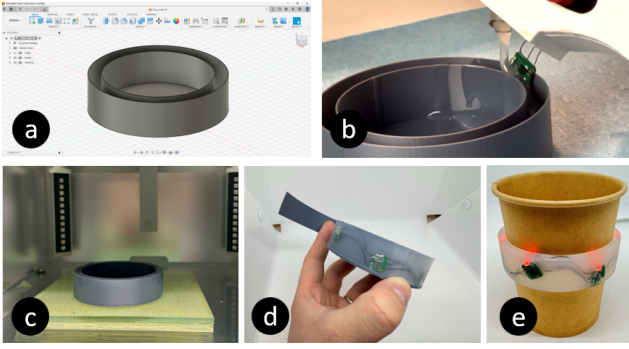


Figure 5: Example manufacturing process of the EmbedNet with silicone rubber (Eco-Flex): (a) Design of the mold. (b) Sensor nodes programming and casting with EmbedNet. (c) Curing. (d) Remove the mold. (e) Connect the sensor strip with a Host, and we then get a function casted object, this example is a simple cup sleeve with EmbedNet sensor nodes to indicate the water temperature inside the paper cup.

application examples spanning different materials and dimensions, ranging from wearables to home sensing.

3.1 Embedding into objects: temperature-sensing cup holder

utilizing EmbedNet, we have designed a casted paper cup holder with integrated sensor nodes. This cup holder features a convenient handle for easy gripping and incorporates sensor nodes to indicate the water temperature inside the paper cup. The RGB LED on each sensor node visually represents the water temperature range. Specifically, blue indicates water temperatures below 20 °C, green signifies temperatures ranging from 20 °C to 45 °C, and red indicates temperatures exceeding 45 °C, suggesting that the water may be too hot for consumption. Figure 6 showcases the functionality of this temperature-sensing cup holder design.

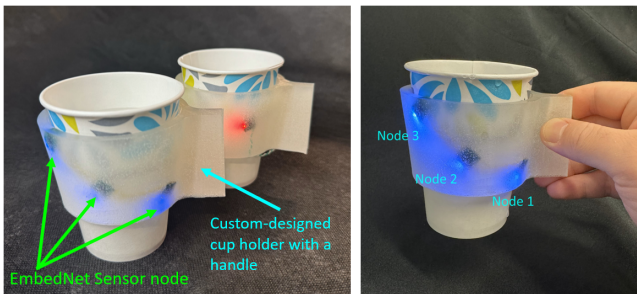


Figure 6: Paper cup holder with EmbedNet sensors inside to indicate water temperature.

The casted mold for the cup holder is crafted from silicone rubber (OOMOO by Smooth-On), while the cup holder itself is fabricated from liquid plastic (Smooth-Cast by Smooth-On).

3.2 Wearable: shoe sole

We introduce a shoe sole equipped with EmbedNet sensor nodes. Leveraging the barometer data from these nodes, we map the pressure distribution on the shoe sole. The shoe sole, crafted from silicone rubber (OOMOO, SMOOTH-ON), houses six EmbedNet sensor nodes to monitor pressure at various locations through the onboard barometer. The OOMOO layer fills the space between the shoe's outer sole and the upper part. Upon curing, the OOMOO effectively bonds these components together. Its flexibility also enables it to function as a damping layer for the shoe. Embedded within are six EmbedNet sensor nodes, strategically positioned to monitor pressure from various locations, providing insights into foot gestures. This setup is depicted in Figure 7.

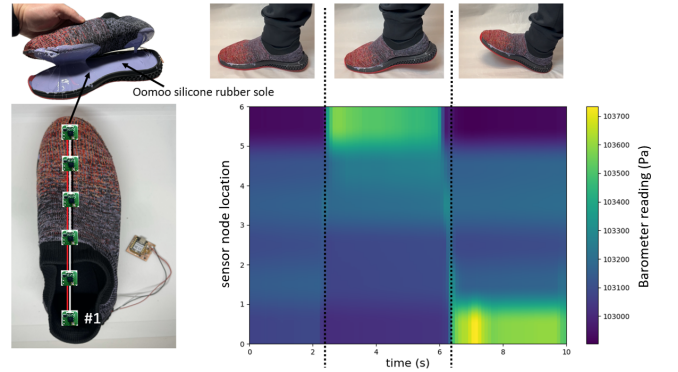


Figure 7: Lab experiment with the custom-made shoe. The barometer data can be used to monitor the foot gesture.

3.3 Wearable: direct-on-body interaction

The versatility of EmbedNet extends to creating wearables directly on the human skin. We utilize Eco-flex Fast (Smooth-On) silicone rubber as a base material, embedding a sensor strip within. This combination is poured directly onto an arm, resulting in a direct-on-body wearable, as depicted in Figure 8.



Figure 8: Making a wearable device by directly applying EmbedNet on the body: (a) Applying EmbedNet on the arm. (b) After curing, the sensor nodes serve as tap buttons. (c) The tap buttons are used to control a snake game.

Equipped with accelerometers capable of detecting tapping, each sensor node serves as a tactile button. Exploiting this feature, we construct an on-body interaction system where users can engage with EmbedNet by tapping on specific nodes to trigger various actions or commands. This intuitive interface facilitates seamless control and interaction with the system.

4 Conclusion

In this paper, we introduce EmbedNet, a system designed to integrate self-sensing capability into casted objects during manufacturing. EmbedNet utilizes custom-designed extruders that are compatible with various base materials for casting purposes. A handheld extruder facilitates the simultaneous extrusion of the base material alongside a sensor strip. Upon pouring this composite mixture into a custom-designed mold and allowing it to cure, we obtain an object with a defined shape and embedded sensing capability. The sensor strip comprises multiple small sensor nodes, each equipped with a microcontroller, RGB LED, and a diverse array of sensors, including accelerometers, Lux sensors, barometers, and temperature sensors. This comprehensive sensor suite caters to a wide range of applications spanning from home sensing to human-computer interaction (HCI) and wearables. With ongoing advancements, EmbedNet can seamlessly integrate sensing capabilities into casted objects, opening up many possibilities for enhanced user experiences and novel applications in fields such as IoT, healthcare, and beyond.

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